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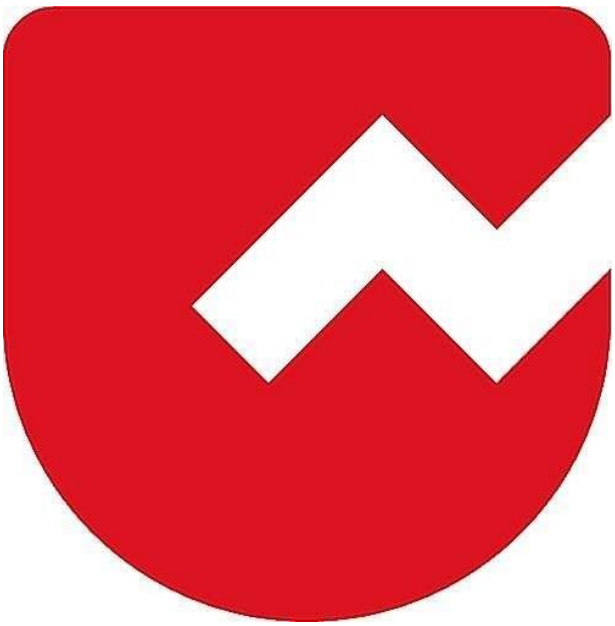


Notes

METHODS

1. What is Method ?

Enclosing set of statement that performs unique task.

2. Syntax of Methods

[access modifier] [non-access modifier] returnType <methodName>([parameterList]) [throws] [Exception list] {

    // set of statements

}

3. Why we need a method or what is the primary purpose of a method?

- **Code Re-usability**: A method needs to be written only once, but it can be invoked multiple times from different places in the program whenever required.

- **Modularity**: Modularity breaking program into parts. Methods divide a large program into smaller, manageable, and organized pieces.

4. What is the advantages of Modularity?

- Easy debugging
- Improves code readability

5. Where generally in the program logic we write?

- All logic and executable statements are written inside methods.
- Business logic resides inside a method.

6. Parameter and Arguments

*MethodDemo.java

```
1 package part1;
2
3 public class MethodDemo {
4     public static void main(String[] args) {
5         go(5); // '5' argument
6     }
7     static void go(int x) // 'x' is a parameter
8     {
9         System.out.println(x);
10    }
11 }
```

7. What is the difference between parameter and arguments

- Parameters are variable(that stores the copy of an arguments)
- Arguments are data or value.

8. How many parameters method can have ?

- N numbers of parameters

9. Variable Length Arguments

```
*MethodDemo.java
 1 package part1;
 2
 3 public class MethodDemo {
 4     public static void main(String...args) { //variable length arguments
 5         go(5); // '5' argument
 6   }
 7     static void go(int x) // 'x' is a parameter
 8     {
 9         System.out.println(x);
10     }
11 }
```

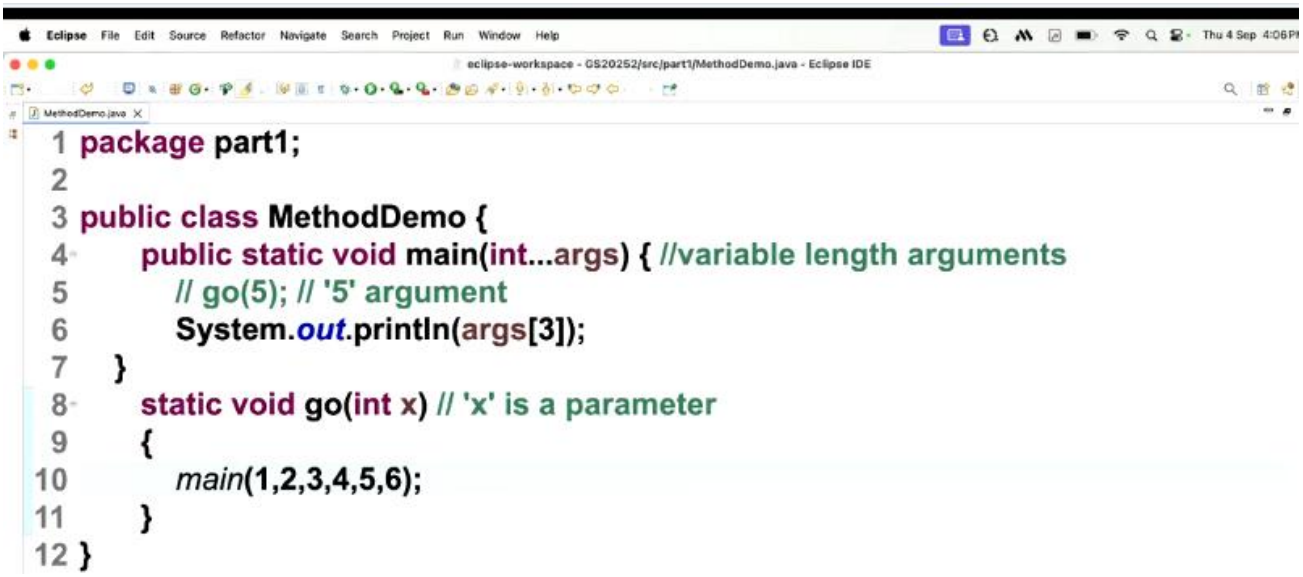
- varargs allow a method to accept zero or more arguments of the same type.

- Instead of fixing the number of parameters, you use ... (three dots) after the type.

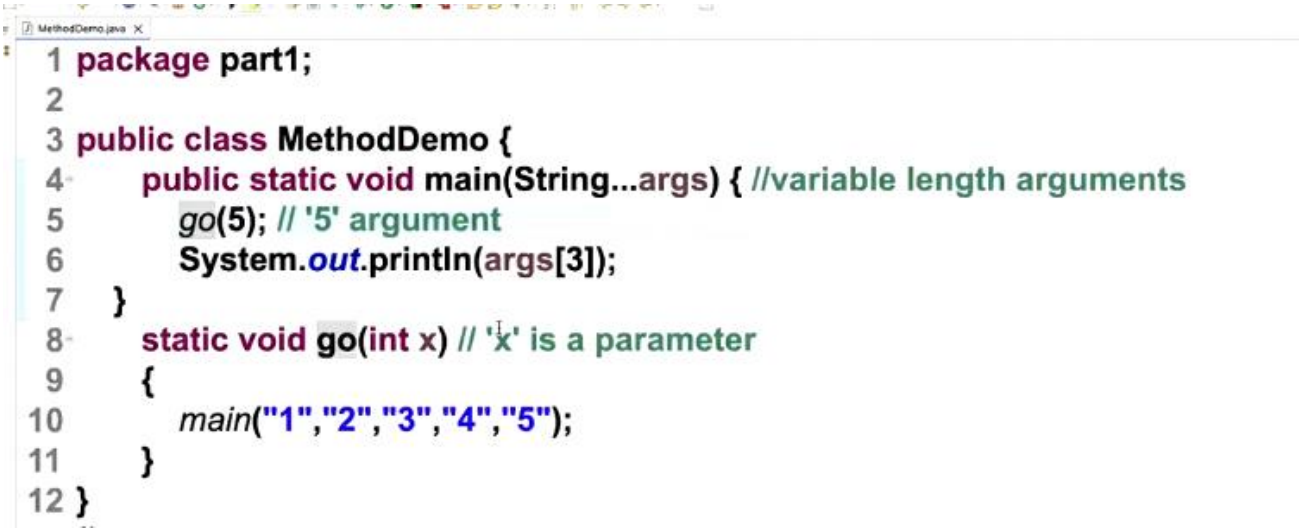
10. Can I call the main method from another method?

- Yes, It is perfectly legal to call the main method from another method. but the method which invoke the main method should be static.

```
MethodDemo.java
 1 package part1;
 2
 3 public class MethodDemo {
 4     public static void main(String...args) { //variable length arguments
 5         // go(5); // '5' argument
 6         System.out.println(args[3]);
 7   }
 8     static void go(int x) // 'x' is a parameter
 9     {
10         main("1","2","3","4","5");
11     }
12 }
```



- Output : Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: 3.
- Because, The statement main("1", "2", "3", "4", "5"); is inside go(int x) but go() is never called.



- Output : Exception in thread "main" java.lang.StackOverflowError
- Because there's no stopping condition, the program will keep calling main and go until the stack overflows.

MethodDemo.java

```
1 package part1;
2
3 public class MethodDemo {
4     public static void main(String...args) { //variable length arguments
5         ho(1,2,3,4,5);
6     }
7     static void  ho(int...a)
8       {
9        System.out.println(a[3]);
10      }
11 }
```

Output : 4

MethodDemo.java

```
1 package part1;
2
3 public class MethodDemo {
4     public static void main(String...args) { //variable length arguments
5         ho(1,2,3,4,5);
6     }
7     static void  ho(int...a)
8       {
9        System.out.println(a.length);
10      }
11 }
```

Output : 5

MethodDemo.java

```
1 package part1;
2
3 public class MethodDemo {
4     public static void main(String...args) { //variable length arguments
5         ho(1,2,3,4);
6     }
7     static void  ho(int b, int...a)
8       {
9        System.out.println(a.length + b);
10      }
11 }
```

Output : 4

*MethodDemo.java

```
1 package part1;
2
3 public class MethodDemo {
4     public static void main(String...args) { //variable length arguments
5         ho(1,2,3,4,5);
6     }
7     static void  ho(int...a,int...b)
8       {
9        System.out.println(a.length + b);
10      }
11 }
```

*MethodDemo.java

```
1 package part1;
2
3 public class MethodDemo {
4     public static void main(String...args) { //variable length arguments
5         ho(1,2,3,4,5);
6     }
7     static void  ho(int...a,int b)
8       {
9        System.out.println(a.length + b);
10      }
11 }
```

These code will not compile.

Because,

RULE 1 : when you use variable-length arguments it **must be the last parameter** in the method signature.

RULE 2 : A method can have at most one varargs parameter in its parentheses.

11. Can I declare a method with default keyword?

Yes can, but the method only appear in the interface, In other words Interface alone can have default methods.

```java
package part1;

public class MethodDemo {
    public static void main(String...args) { //variable length arguments
        ho(1,2,3,4,5);
    }
    static String  ho(int...a)
    {
     System.out.println(a.length);
     return ;
    }
}
```

Output : the code will not compile.
Because, Null is not a String.

If you want to return a empty string

```java
package part1;

public class MethodDemo {
    public static void main(String...args) { //variable length arguments
        ho(1,2,3,4,5);
    }
    static String  ho(int...a)
    {
     System.out.println(a.length);
     return "";
    }
}
```

```java
public class MethodDemo {
    public static void main(String...args) { //variable length arguments
        ho(1,2,3,4,5);
    }
    static String  ho(int...a)
    {
     System.out.println(a.length);
     return null;
    }
}
```

- It is perfectly legal for a method to return `null`.
- `NULL -> Means Undefined or Unknown`

12. Can I interchange access modifier and non-access modifier ?

- Non-access modifier can appear before Access modifier.